

Thermoelectric Properties of Authentic Aerogels Incorporating PEDOT:PSS with Mesoporous Structures Fabricated by Supercritical Drying

Haruka Goto, Naoya Yanagishima, Mone Terasawa, Shinji Kanehashi, Kenji Ogino, and Takeshi Shimomura*



Cite This: *ACS Appl. Polym. Mater.* 2025, 7, 426–433



Read Online

ACCESS |



Metrics & More



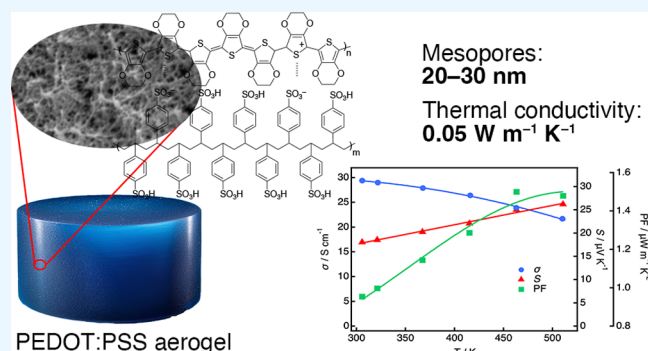
Article Recommendations



Supporting Information

ABSTRACT: We focused on clarifying the structure, thermal and electric conducting properties, and thermoelectric (TE) properties of an authentic aerogel comprising poly(3,4-ethyrenedioxythiophene):poly(styrenesulfonate) (PEDOT:PSS) that was fabricated via supercritical CO₂ drying of an alcogel of cross-linked PEDOT:PSS. Most studies on porous organic TE materials have focused on cryogels with macroporous structures; few reports exist on authentic aerogels with mesoporous structures that exhibit TE effects. The fabricated PEDOT:PSS aerogel had a large surface area (300 m² g^{−1}) and mesopores (20–30 nm), which effectively restricted gas molecule convection, resulting in a low thermal conductivity of approximately 0.05 W m^{−1} K^{−1}. Moreover, this unique structural integrity was retained up to nearly 300 °C. The conductivity was significantly enhanced by cross-linking treatment with divinyl sulfone (DVS) and methanol replacement prior to supercritical CO₂ drying, yielding an effect similar to that of secondary doping. The treated PEDOT:PSS aerogel exhibited a negative temperature dependence on conductivity, which can be attributed to the band-like carrier conduction typically observed in degenerate semiconductors. The relationship between the conductivity and Seebeck coefficient as a function of the temperature is also consistent with the behavior of degenerate semiconductors. Consequently, the PEDOT:PSS aerogel can be regarded as a mesoporous TE material with low thermal conductivity and the characteristic properties of a degenerate semiconductor.

KEYWORDS: conducting polymer, aerogel, porous material, thermoelectric property, PEDOT:PSS, degenerate semiconductor



INTRODUCTION

Thermoelectric (TE) materials with microporous or nanoporous structures have been investigated extensively because of their low thermal conductivity owing to their high air content, which maintains the temperature difference and enhances the TE performance. Among the various types of porous materials, xerogels, cryogels, and aerogels have attracted attention because of their simple fabrication process and high thermal insulation properties.¹ In particular, aerogels,^{2,3} which are fabricated by the supercritical fluid (typically CO₂) drying of hydro/alcogels that were synthesized by condensation, sol–gel methods, or polymer cross-linking, have characteristic mesopores ranging from a few to tens of nanometers by avoiding shrinkage due to the surface tension of the solvent during the drying process. These mesopores, which are smaller than the mean free path of gas molecules, lead to lower thermal conductivity than that of ambient air based on the Knudsen effect.⁴ An aerogel structure that can preserve a temperature difference should be suitable for TE devices.

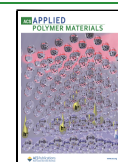
For inorganic materials, a few reports on authentic aerogels, excluding cryogels, that display TE effects have been published. An aerogel comprising Bi₂Te₃ and Bi_xSb_{2−x}Te₃ nanoparticles fabricated using the sol–gel method and supercritical CO₂ drying has been reported to exhibit the TE effect and has considerably low thermal conductivity and $ZT = 0.003$.⁵ Furthermore, composites with silicate aerogel and Bi₂Te₃ (Bi_xSb_{2−x}Te₃)-based materials enhanced ZT to 1.12 (425 K).⁶ In contrast, few reports exist on authentic aerogels that exhibit the TE effect in organic and polymeric materials, and almost all reports on TE porous materials described cryogels. Polymeric materials have unique characteristics that are impossible to realize in inorganic materials and are commonly applied as heat

Received: October 24, 2024

Revised: December 11, 2024

Accepted: December 12, 2024

Published: December 23, 2024



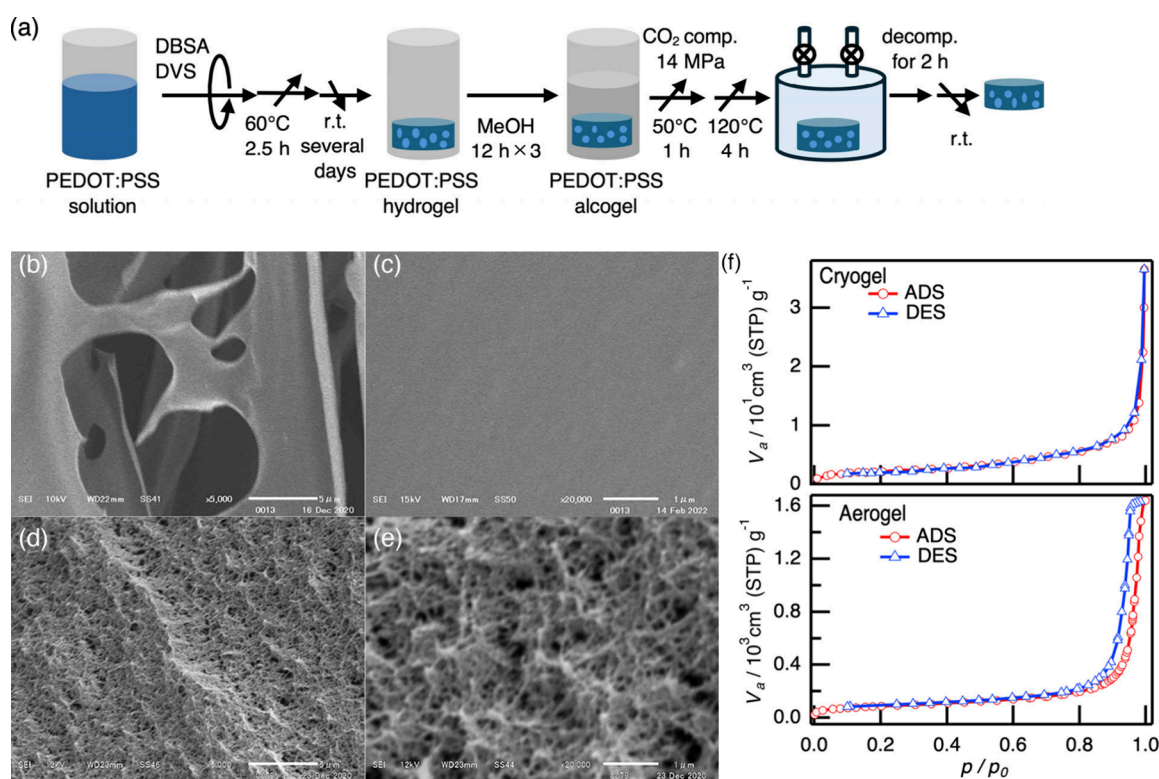


Figure 1. (a) Schematic image of the fabrication of the PEDOT:PSS aerogel. SEM images of (b) PEDOT:PSS cryogel ($\times 5,000$), (c) PEDOT:PSS cryogel under high magnification ($\times 20,000$), (d) PEDOT:PSS aerogel (DVS = 4.0 vol %, $\times 5,000$), and (e) PEDOT:PSS aerogel under high magnification (DVS = 4.0 vol %, $\times 20,000$). (f) Adsorption–desorption isotherm of PEDOT:PSS cryogel and aerogel (DVS = 4.0 vol %). Open circles (red) represent the adsorption profile, and open triangles (blue) represent the desorption profile.

insulators, as exemplified by polystyrene or polyurethane foams. Recently, poly(3,4-ethylenedioxythiophene):poly(styrenesulfonate) (PEDOT:PSS) cryogels have been applied in TE devices.^{7–13} Urban et al. fabricated a PEDOT:PSS cryogel by freeze-drying hydrogel from a thick film, excluding additional cross-linking agents, and investigated its TE effect.⁷ We also fabricated PEDOT:PSS cryogels and progressed the secondary doping method without breaking the porous structure using a supercritical CO₂ atmosphere.⁸ Naguib et al. reported freeze-drying 3D-printed hydrogels into cryogels and described their thermoelectric properties.⁹ Cryogels blended with PEDOT:PSS and other materials have been reported, including those blended with nanofibrillated cellulose,^{10,11} CNTs, and Ag¹² and Te nanowires.¹³ In addition to those of PEDOT:PSS, the thermoelectric properties of cryogels of poly(3-hexylthiophene) blended with other polymers have been reported.^{14–16} Although these materials were termed “aerogels” in the papers, they were cryogels fabricated via freeze-drying and did not have the mesopores characteristic of aerogels. Very recently, Biniek et al. reported an authentic aerogel of PEDOT:PSS and described its thermoelectric properties.¹⁷ This aerogel was fabricated via supercritical CO₂ drying of an alcogel (without chemical cross-linking) and had mesopores ranging between 20 and 50 nm and a large surface area of $315 \text{ m}^2 \text{ g}^{-1}$, comparable to that of silica aerogels, unlike the cryogel.

In this study, we aimed to clarify the poorly understood structures, thermal and electrical conducting properties, and TE properties of authentic aerogels containing PEDOT:PSS. In particular, aerogel synthesis conditions, such as cross-linking methods and cross-linking density, should affect the thermal/electric conducting and TE properties. Therefore, we fabricated

PEDOT:PSS aerogels with extremely low thermal conductivity and focused on the carrier transport mechanism, which is strongly related to the TE properties. The carrier transport of conducting polymers is acknowledged as being dominated by hopping transports, including nearest neighbor and variable range hopping, which exhibit a thermal activation-type temperature dependence characteristic to a nondegenerate semiconductor even at a high doping region in many cases.¹⁸ However, in PEDOT:PSS treated with adequate secondary doping, several reports describe the conductivity as being almost independent of temperature or describe the conductivity as having a negative temperature coefficient, which could be explained as fluctuation-induced tunneling or band-like conduction (metal-like conduction) at approximately room temperature.^{19–22} Here, we aimed to clarify whether these band-like conduction could also be observed in the aerogel structure of PEDOT:PSS. The appearance of band-like conduction should significantly expand the range of applications of PEDOT:PSS aerogels.

EXPERIMENTAL METHODS

To prepare the PEDOT:PSS aerogel, a PEDOT:PSS aqueous dispersion (1.0–1.3 wt %, Clevios PH1000) was purchased from Heraeus Electro-Nite Japan Ltd. Divinyl sulfone (DVS) and dodecylbenzenesulfonate (DBSA) were purchased from Tokyo Chemical Industry Co., Ltd. and used without further purification. The synthesis of a PEDOT:PSS hydrogel followed a method described previously,²³ and cross-linking mechanisms have been suggested elsewhere (Figure S1).^{7,24} DBSA (0.5 vol %) (physical cross-linker) and DVS (1.0–10.0 vol %) (chemical cross-linker) were added to 5 mL of PEDOT:PSS aqueous dispersion, and the mixture was stirred for 5 min. After the mixture was heated at 60 °C for 2.5 h and maintained

undisturbed for several days at 25 °C, a PEDOT:PSS hydrogel was obtained. When a base, such as triethylamine, was added during this reaction, gelation was accelerated but dedoping of PEDOT:PSS occurred. Therefore, we performed gelation under acidic conditions originating from the PEDOT:PSS. The hydrogel was dipped three times in pure water, twice in a mixture of pure water and methanol (50:50), and three times in methanol for 12 h. Thereafter, an alcogel in which water was almost completely exchanged with methanol was obtained.

The alcogel, contained in a lab dish, was placed in a pressure-resistant vessel; CO₂ was injected into the vessel to be pressurized to 14 MPa at 50 °C and maintained for an hour under this condition in which CO₂ was in a supercritical state. After heating the vessel at 120 °C and maintaining this condition for 4 h, decompression was performed slowly for 2 h, and the sample was gradually cooled to 25 °C. Finally, an aerogel was obtained. For supercritical drying, a CO₂ pump (SCF-Get, JASCO Co.), a heat oven (RO-969, JASCO Co.), and a back-pressure regulator (SCF-Bpg, JASCO Co.) were used. The mass, area, and thickness of the aerogels were measured after treatment.

A cryogel of PEDOT:PSS, fabricated without cross-linker for the control, was prepared according to the method described in a previous report.⁸ PEDOT:PSS aqueous dispersion (4 mL) was preliminarily frozen in liquid N₂ and freeze-dried overnight (DC801, Yamato Scientific Co., Ltd.).

For characterization of the PEDOT:PSS aerogel, scanning electron microscopy (SEM) was conducted using a JEOL JSM-6510 microscope. Wide-angle X-ray diffraction (WAXD; NANO-Viewer system, Rigaku Co., Ltd.) measurements were performed using Cu K α radiation with a wavelength of 1.54 Å at 46 kV and 60 mA. An imaging plate (IP) (BAS-SR 127, Fujifilm Corp.) was used as a two-dimensional detector, and an IP reading device (RAXIA-Di, Rigaku Co., Ltd.) was used to transform the obtained images into text data. Thermogravimetric analysis (TGA) was performed using a thermogravimetric analyzer (TGA-50, Shimadzu Co.) by flowing drying nitrogen gas at a heating rate of 10 °C/min in the temperature range between 25 and 900 °C. A uniaxial compression-tensile testing machine (EZ-SX, Shimadzu) was used to investigate the strength of the PEDOT:PSS aerogels.

Nitrogen sorption measurements were performed to obtain the Brunauer–Emmett–Teller (BET) specific surface area and pore size distribution of the PEDOT:PSS aerogel by using a BET surface area analyzer (BELSORP-max, MicrotracBEL Co.). The specific surface areas of the samples were determined from the linear portion of the BET plots, and the total pore volume was determined from the amount adsorbed at a relative pressure of 0.990. The pore size distribution was calculated from the adsorption or desorption isotherms by using the conventional Barrett–Joyner–Halenda (BJH) method.

Thermoelectric measurements were performed using a Seebeck coefficient/electric resistance measurement system (ZEM-3; Advance Riko Inc.). The electrical conductivity, σ , was measured using the 4-probe method, and the Seebeck coefficient, S , was calculated from $V = S\Delta T$, where V is the induced voltage and ΔT is the temperature gradient in a helium atmosphere. Thermal diffusivity α was measured by both a laser flash method (LFA-502, Kyoto Electronics Manufacturing Co., Ltd.) and a Fourier transform thermal wave analyzer (ai-Phase Mobile 1u, ai-Phase Co., Ltd.). Thermal conductivity was estimated as $\kappa = \alpha C_p \rho$, where C_p and ρ denote the specific heat capacity and density of the aerogel, respectively.

RESULTS AND DISCUSSION

Characterization of PEDOT:PSS Aerogel Structure. The procedure used to prepare the aerogels is shown in Figure 1a. PEDOT:PSS took 2–6 days to gel upon the addition of DVS and DBSA as cross-linkers (Figure S2). The PEDOT:PSS aerogel was obtained by supercritical drying of a PEDOT:PSS alcogel after solvent replacement from water. The bulk density ρ and porosity p of the PEDOT:PSS film (fabricated by casting without cross-linking), cryogel (fabricated by freeze-drying without cross-linking), and aerogel are listed in Table 1.

Table 1. Bulk Density (ρ) and Porosity (p) of the PEDOT:PSS Film, Cryogel, and Aerogel Treated with Different Amounts of DVS

	DVS/vol %	$\rho_{\text{bulk}}/\text{g cm}^{-3}$	$p/\%$
Film	0	1.36	
Cryogel	0	1.65×10^{-2}	98.9
Aerogel	1.0	0.112 ± 0.038	90.3 ± 3.3
Aerogel	2.0	0.101 ± 0.029	91.3 ± 2.5
Aerogel	4.0	0.104 ± 0.022	91.0 ± 2.0
Aerogel	10.0	0.144 ± 0.017	87.5 ± 1.5

PEDOT:PSS aerogels, which were synthesized using a cross-linker, had a density that was an order of magnitude higher than that of the PEDOT:PSS cryogels fabricated without a cross-linker. Nevertheless, all aerogels featured a p of approximately 90%, indicating a low density.

SEM images of frozen sections of both the PEDOT:PSS cryogel and aerogel (DVS = 4.0 vol %) were obtained. The cryogel had an entangled sheet structure with relatively large pores, approximately 10 μm in size, as shown in Figure 1b. A flat surface devoid of mesopores was observed in the high-magnification image of the cryogel (Figure 1c). Conversely, the PEDOT:PSS aerogel had a structure significantly denser than that of the cryogel, which is consistent with the bulk density measurements shown in Figure 1d. The SEM images of the aerogels prepared under different conditions are shown in Figure S3. In the high-magnification image, the mesopores, which are characteristic of inherent aerogels, were evenly distributed, as shown in Figure 1e. Thus, a genuine aerogel structure with mesopores can only be achieved via supercritical drying of the alcogel in the case of PEDOT:PSS.

As the SEM analysis confirmed the presence of mesopores in the PEDOT:PSS aerogel, the surface area and pore size distribution were examined by using a BET surface area analyzer. The PEDOT:PSS cryogel exhibited an adsorption–desorption isotherm characteristic of materials with macropores or lacking pores (type II), as illustrated in Figure 1f. In contrast, the isotherm of the PEDOT:PSS aerogel displayed hysteresis and was classified as a mesoporous material (type IV), as illustrated in Figure 1f. These results were consistent with the SEM observations. The isotherms and BET plots of the PEDOT:PSS aerogels that were prepared under different conditions are presented in Figures S4 and S5, respectively.

The specific surface area a_s estimated by the BET plot, total pore volume V , and mode pore size r_p estimated using the BJH method (Figure S6 and Table S1) are listed in Table 2. The a_s of the PEDOT:PSS aerogel was 37–58 times higher than that of the PEDOT:PSS cryogel and tended to decrease with increasing DVS content. Furthermore, a_s exceeding 100 $\text{m}^2 \text{g}^{-1}$ has not

Table 2. Specific Surface Area (a_s), Total Pore Volume (V), and Mode Pore Size (r_p) of PEDOT:PSS Cryogel and Aerogel Treated with Different Amounts of DVS^a

	DVS/vol %	$a_s/\text{m}^2 \text{g}^{-1}$	$V/\text{cm}^3 \text{g}^{-1}$	r_p/nm
Cryogel	0	7.71		
Aerogel	1.0	378	2.15	29.1
Aerogel	2.0	445	3.75	29.6
Aerogel	4.0	346	2.06	18.0
Aerogel	10.0	283	1.57	16.6

^aAverage values.

been previously reported for PEDOT:PSS cryogels.²⁵ The PEDOT:PSS aerogel, which exhibited $a_s \geq 300 \text{ m}^2 \text{ g}^{-1}$ and is comparable to that of conventional silica aerogels, was unique and confirmed to be a genuine aerogel. The distinctive properties of mesopores with a r_p ranging between 15 and 30 nm were not observed in the PEDOT:PSS cryogel.²⁵

WAXD measurements were conducted to investigate the microstructure of the PEDOT:PSS aerogel. Figure 2a shows the

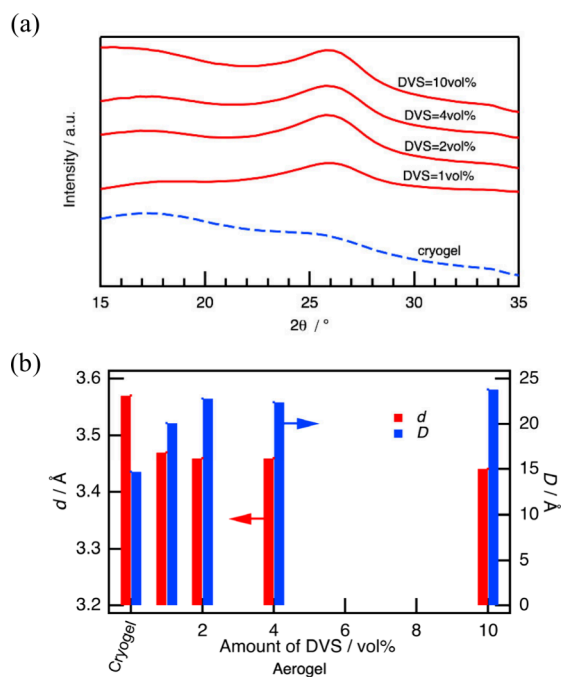


Figure 2. (a) WAXD patterns for the PEDOT:PSS cryogel and aerogels upon changing the amount of DVS. (b) d -Spacing (red bar) and crystallite diameter D (blue bar) of the (020) peak for the PEDOT:PSS cryogel and aerogel treated with different amounts of DVS.

WAXD patterns of the PEDOT:PSS cryogel and aerogel. On the PEDOT:PSS cryogel, the characteristic peak at approximately $2\theta = 25^\circ$, which corresponds to the interchain distance of the π - π stack direction (020) of PEDOT, was obscure, because the cryogel was not treated with a secondary doping and the crystallinity might be low. In contrast, for the PEDOT:PSS aerogel, a peak appeared clearly at approximately $2\theta = 25^\circ$, regardless of the DVS concentration. On the aerogel, an effect similar to secondary doping was observed because cross-linking was performed and the aerogel was prepared via an alcogel before supercritical drying. The d -spacing and crystallite diameter, D , of the (020) peak, estimated using the Scherrer equation, are illustrated in Figure 2b. In the aerogel, the d -spacing was smaller and D was larger than that of the cryogel. Given these results, the procedure for preparing the aerogel evidently promoted crystallization, which had an effect similar to that of secondary doping and was advantageous for the electronic properties of the aerogels.

Figure 3 shows the stress–strain curves of the PEDOT:PSS aerogels comprising different amounts of DVS. The cryogel without cross-linking could not be stretched, whereas the aerogels cross-linked with DBSA and DVS could be stretched by tens of percent. Aerogels, including silica aerogels, are acknowledged as being quite fragile; these PEDOT:PSS aerogels belong to the strong category of aerogels. The Young's modulus was approximately 10–20 MPa (Table S2) and increased with

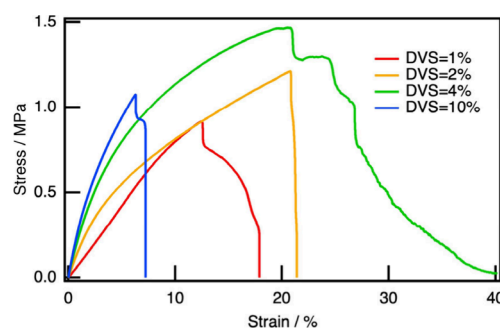


Figure 3. Stress–strain curve of PEDOT:PSS aerogels comprising different amounts of DVS.

increasing amounts of DVS; furthermore, the Young's modulus was an order of magnitude larger than that of the silica aerogel. These characteristic mechanical properties of the PEDOT:PSS aerogels were ascribed to the cross-linking between the PEDOT:PSS molecules.

The TGA charts of the PEDOT:PSS film and PEDOT:PSS aerogel (DVS = 4.0 vol %) in Figure 4 show three weight-loss

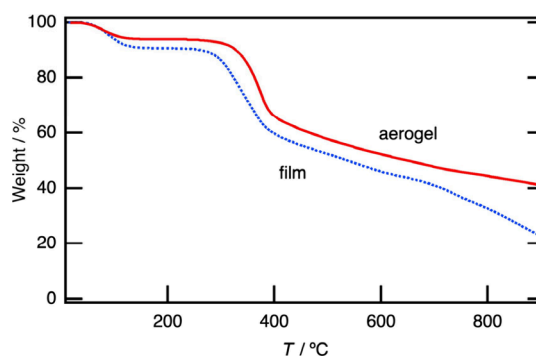


Figure 4. TGA chart of PEDOT:PSS film and PEDOT:PSS aerogel (DVS = 4.0 vol %).

steps. The first step is water loss. The second step at approximately 300 °C corresponds to decomposition of PSS via the rupture of the sulfonic group from styrene, and the broad third step above 400 °C corresponds to decomposition of the main chain of PEDOT and PSS.²⁶ The second step observed at 293 °C in the film shifted to 335 °C in the aerogel, and this shift implies that the aerogels have been cross-linked by DVS and DBSA. The PEDOT:PSS aerogel could maintain its structure up to high temperatures exceeding 300 °C. Furthermore, the SEM image of the aerogel annealed at 250 °C revealed that the mesoporous structure was maintained, as shown in Figure S7.

Thermal Conductivity of PEDOT:PSS Aerogel. Thermal diffusivity α was measured by laser flash and Fourier transform thermal wave analyzing methods (Figure 5). In the aerogels comprising a low amount of DVS (1.0 and 2.0 vol %), α , measured by thermal wave analysis, was approximately twice as large as that measured by the laser flash method. For porous materials, even if the pore size is sufficiently small to restrict the diffusion of gas molecules, the effect of gas convection may be included in the total diffusivity using thermal wave analysis in which periodic heat is continuously applied to the sample. In contrast, in the laser flash method, which can instantaneously apply heat, the effect of gas convection is less likely to occur, and the thermal diffusivity of the material component can be primarily measured. As the difference between the two methods

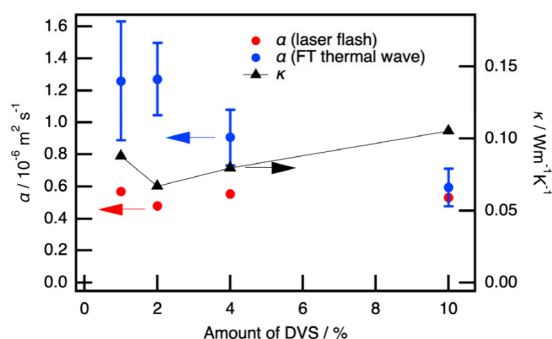


Figure 5. Thermal diffusivity α measured by the laser flash (red circles), Fourier transform thermal wave analyzing methods (blue circles), and thermal conductivity κ (black triangles).

decreased with an increase in the amount of DVS, gas convection was almost completely suppressed when the DVS amounts were 4.0 and 10.0 vol % with a pore radius of less than 20 nm. Notably, the aerogels exhibited an extremely low α of approximately $5 \times 10^{-7} \text{ m}^2 \text{ s}^{-1}$ under all conditions during analysis by the laser flash method. The thermal conductivity κ was estimated (Figure 5) from the α obtained by the laser flash method, specific heat capacity at a constant pressure of PEDOT:PSS C_p ($=1.38 \text{ J g}^{-1} \text{ K}^{-1}$), and density ρ . This aerogel had a comparably low thermal conductivity to that of previously reported cryogels comprising PEDOT which have a κ value of the order of 8.0×10^{-2} – $1.4 \times 10^{-1} \text{ W m}^{-1} \text{ K}^{-1}$.^{1,9,13}

Thermoelectric Properties of PEDOT:PSS Aerogel.

Figure 6a presents the electrical conductivity (σ) and Seebeck coefficient (S) at 300 K of PEDOT:PSS cryogels and aerogels comprising different amounts of DVS. The PEDOT:PSS aerogel had a markedly higher σ ($\sim 35 \text{ S cm}^{-1}$) than that of the PEDOT:PSS cryogel (0.078 S cm^{-1}), corresponding to an approximately 450 times increase. This significant increase in σ was consistent with the reduction of the d -spacing of the π - π stacking; moreover, the increase in the crystallite sizes of PEDOT resulted from WAXD.

As these conductivities σ were the apparent conductivity of the porous structure that had a large air volume, the intrinsic conductivity of the PEDOT:PSS material should be larger than these measured values. Therefore, the intrinsic conductivity σ_0 of the PEDOT:PSS material was estimated using the following equation²⁷ and is shown in Figure 6a.

$$\sigma_0 = \frac{\sigma}{\{(1-p) + 2(1-p)^{3/2}\}/3} \quad (1)$$

The intrinsic conductivity σ_0 of the PEDOT:PSS aerogel, estimated to be 650 S cm^{-1} , was significantly larger than that of the PEDOT:PSS cryogel (18 S cm^{-1}); therefore, this advantage was not due to the density of the aerogels being higher than that of the cryogels. Furthermore, the magnitude of σ_0 of the PEDOT:PSS aerogels was similar to that of secondary doped PEDOT:PSS.¹⁹

The conductivity of the PEDOT:PSS film (not aerogel) treated by dipping in methanol and/or cross-linking by DBSA/DVS was investigated (Figure 6b) to confirm an effective protocol to increase σ during fabrication of the aerogel. Dipping the PEDOT:PSS film in methanol resulted in a marked increase in conductivity that was similar to that which resulted from secondary doping by a high boiling-point solvent as reported previously.⁸ In contrast, the cross-linking treatment also increased the conductivity; in particular, cross-linking by DVS

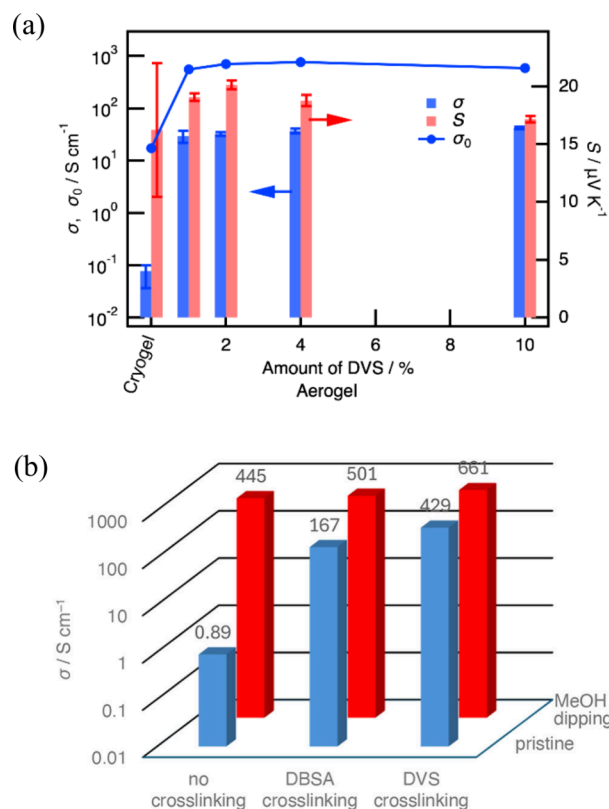


Figure 6. (a) Conductivity (σ) and Seebeck coefficient (S) of the PEDOT:PSS cryogel and aerogels comprising different amounts of DVS. σ and σ_0 are the apparent conductivity of the cryo/aerogel structure and intrinsic conductivity of the PEDOT:PSS material in the cryo/aerogel, respectively. (b) σ of the PEDOT:PSS film treated by dipping in methanol and/or cross-linking by DBSA/DVS.

resulted in a marked increase in the conductivity which was of the same extent as that caused by the methanol dipping treatment. Both treatments appeared to contribute to the destruction of the PSS shell structure of the PEDOT:PSS colloids and improved the crystallinity of the PEDOT core structure. When both processes are used, a higher conductivity could be obtained synergistically.

In contrast, the S values of the aerogels were similar to those of the cryogels, as shown in Figure 6a. The Seebeck coefficient for PEDOT:PSS is reported to range between 10 and $30 \mu\text{V K}^{-1}$,^{7–9} and the result obtained in this study ($\sim 20 \mu\text{V K}^{-1}$) is consistent with the reported values. Hence, aerogelation has no significant effect on the Seebeck coefficient. The power factor, PF ($= S^2 \sigma$), of the aerogel was approximately $1.0 \mu\text{W m}^{-1} \text{ K}^{-2}$, larger by approximately 4 orders of magnitude compared with that of the cryogel without cross-linking, which was predominantly due to the improvement in conductivity. The thermoelectric figure of merit ZT was estimated to be $\leq 5.96 \times 10^{-3}$ (DVS = 2.0 vol %), which is comparable to the previously reported values for pure PEDOT:PSS cryogel.^{7–9}

The temperature dependence of σ and S was investigated, as it was confirmed that the aerogels were able to maintain their unique structure at high temperatures exceeding 300°C by TGA and SEM measurement. Figure 7a shows the σ , S , and PF of the PEDOT:PSS aerogel (DVS = 4.0 vol %) with increasing temperature from 306 to 511 K. σ decreased slightly with increasing temperature over the entire temperature range measured in this study; the observed change was different

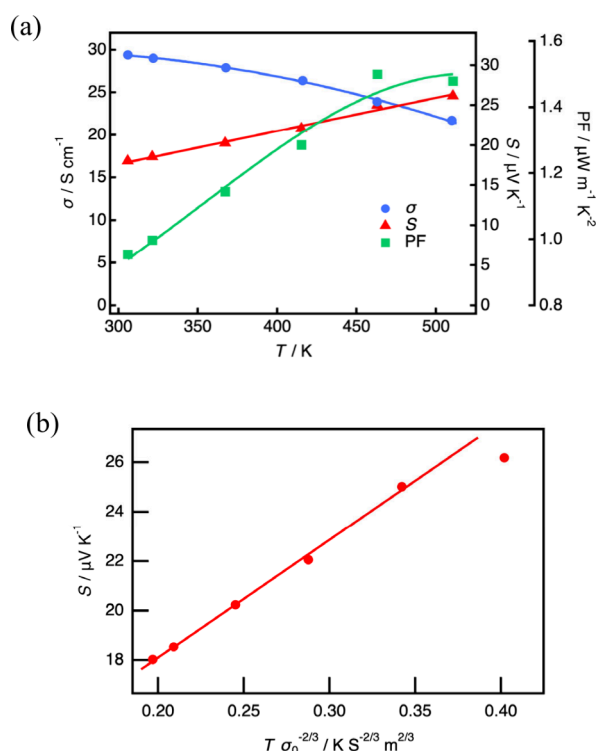


Figure 7. (a) Temperature dependence of σ , S , and PF of PEDOT:PSS aerogel (DVS = 4.0 vol %). (b) Relationship between S and $T/\sigma_0^{2/3}$, where the conductivity adopted is the intrinsic conductivity calibrated by the porosity of the aerogel.

from the dependence of the thermally activated (nondegenerate semiconductor) PEDOT:PSS film that did not undergo secondary doping. Previous reports have indicated that the temperature dependence tends to be subtle after secondary doping, because the carrier transport mechanism changes from variable-range hopping to fluctuation-induced tunneling, thus yielding a metallic state.¹⁹ The negative temperature dependence of σ , observed after secondary doping by methanol dipping and cross-linking by DBSA/DVS, in this study, also appears to be similar to a band-like transport of the degenerate semiconductor similar to that displayed by a few previously reported PEDOT materials.^{21,22}

In contrast, S increased almost linearly with the temperature (Figure 7a). The change in S can be explained by a semiempirical equation as $S \sim T$ in the conducting polymer;²⁸ therefore, an almost linear relationship between S and T is acceptable. Here, in a degenerate semiconductor, the relationship between S and σ may be described by the following equation:^{29,30}

$$S = \frac{k_B}{e} \left(\frac{3}{2} + \gamma \right) \frac{2m^*k_B(\pi e\mu)^{2/3}}{3^{5/3}\hbar^2} \frac{T}{\sigma^{2/3}} \quad (2)$$

where k_B is the Boltzmann constant, e is the elementary charge, γ is the scattering factor, m^* is the effective mass, $\hbar$ is the reduced Plank constant, and μ is carrier mobility. Therefore, a relationship between S and $T/\sigma_0^{2/3}$ is presented in Figure 7b, where the conductivity adopted is the intrinsic conductivity, σ_0 , calibrated by the porosity p of the aerogel because a porous structure is not considered in eq 2. The relationship between S and $T/\sigma_0^{2/3}$ is almost linear, and the slope is estimated to be 4.8×10^{-5} V S^{2/3} m^{-2/3} K⁻¹. When γ is negligible and m^* and μ are assumed to be that of the free electron (m_e) and 1.0 cm² V⁻¹ s⁻¹, respectively, the slope is calculated as 6.4×10^{-5} V S^{2/3} m^{-2/3}

K⁻¹ from eq 2. The experimental result was in good agreement with this value. Therefore, because the relationship between σ and S can be consistently explained by this model, the PEDOT:PSS aerogel is a degenerate semiconductor. Furthermore, in this aerogel, the maximum PF is estimated to be 1.50μ W m⁻¹ K⁻² at 463 K, which is among the higher values obtained for pure PEDOT:PSS cryogels.^{7–9}

PEDOT:PSS and its composite materials have been investigated as TE materials in thin film form,³¹ and some reports exhibited competitive conversion efficiencies.^{21,32} However, the thermal conductivity of PEDOT:PSS (0.2 – 0.37 W m⁻¹ K⁻¹) and its composite materials (0.2 – 0.9 W m⁻¹ K⁻¹)³¹ is relatively high for polymeric materials, and so, it is challenging to maintain the temperature difference in the thin-film configuration. In contrast, PEDOT:PSS authentic aerogels, with thermal conductivity below 0.1 W m⁻¹ K⁻¹, offer a distinctive advantage for a variety of practical applications.

Several conducting polymer cryogels with thermal conductivities comparable to the PEDOT:PSS aerogel in this study have been reported.^{8,9,13–15} However, these cryogels typically feature macropores, leading to additional heat transfer via gas convection, alongside the intrinsic thermal conduction of the material in practical applications. The PEDOT:PSS aerogel presented in this study is the first polymeric TE porous material in which it has been confirmed that gas convection has been effectively suppressed. Furthermore, as this aerogel is composed solely of PEDOT:PSS, the density of the conductive pathway was significantly higher compared to cryogels comprising conducting polymer composites such as P3HT-based cryogels.^{14,15} This structural configuration results in remarkably high electrical conductivity for a porous conducting polymer and exhibits carrier transport characteristics akin to those of degenerate semiconductors typically observed in bulk PEDOT films.

CONCLUSIONS

Authentic PEDOT:PSS aerogels were fabricated by drying alcogels using supercritical CO₂; the alcogels were formed from cross-linked PEDOT:PSS. The resulting PEDOT:PSS aerogels had a large surface area exceeding 300 m² g⁻¹ and mesopores ranging between 20 and 30 nm and demonstrated exceptionally low thermal conductivity, approximately 0.05 W m⁻¹ K⁻¹. The chemical cross-linking and alcogelation treatments markedly improved the conductivity by a factor of 450 . The electrical conductivity and Seebeck coefficient of the PEDOT:PSS aerogel were approximately 35 S cm⁻¹ and 20 μ V K⁻¹, respectively. Furthermore, the conductivity of the PEDOT:PSS aerogel displayed a negative temperature dependence, which was attributed to the band-like carrier transport of degenerate semiconductors. The relationship between the conductivity and Seebeck coefficient as a function of temperature was also consistent with the degenerate semiconductor behavior. The PEDOT:PSS aerogel is a unique class of mesoporous thermoelectric material characterized by low thermal conductivity and degenerate semiconductor properties, suggesting its potential for innovative applications.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsapm.4c03383>.

Cross-link structure of PEDOT:PSS by DVS and DBSA, gelation of PEDOT:PSS by DVS and DBSA, SEM images of PEDOT:PSS aerogel, the adsorption–desorption isotherm of PEDOT:PSS aerogel, porous structure of PEDOT:PSS aerogel annealed at 250 °C, and mechanical properties of PEDOT:PSS aerogel (PDF)

AUTHOR INFORMATION

Corresponding Author

Takeshi Shimomura — Graduate School of Engineering, Tokyo University of Agriculture and Technology, Koganei, Tokyo 184-8588, Japan; orcid.org/0000-0001-9580-1019; Email: simo@cc.tuat.ac.jp

Authors

Haruka Goto — Graduate School of Engineering, Tokyo University of Agriculture and Technology, Koganei, Tokyo 184-8588, Japan

Naoya Yanagishima — Graduate School of Engineering, Tokyo University of Agriculture and Technology, Koganei, Tokyo 184-8588, Japan

Mone Terasawa — Graduate School of Engineering, Tokyo University of Agriculture and Technology, Koganei, Tokyo 184-8588, Japan

Shinji Kanehashi — Graduate School of Engineering, Tokyo University of Agriculture and Technology, Koganei, Tokyo 184-8588, Japan

Kenji Ogino — Graduate School of Bio-Applications and Systems Engineering, Tokyo University of Agriculture and Technology, Koganei, Tokyo 184-8588, Japan

Complete contact information is available at:
<https://pubs.acs.org/10.1021/acsapm.4c03383>

Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

This study was partially supported by KAKENHI (grant nos. JP21K18995 and JP22H02136) from the Japan Society for the Promotion of Science (JSPS). The authors thank Hidehiro Kamiya at TUAT for the BET surface area analysis and Hiromu Saito at TUAT for the WAXD measurement.

REFERENCES

- (1) Radouane, N. Review on Thermoelectric Aerogels and Their Applications: Progress and Challenges. *J. Sol-Gel Sci. Technol.* **2023**, *106*, 639–653.
- (2) Gesser, H. D.; Goswami, P. C. Aerogels and Related Porous Materials. *Chem. Rev.* **1989**, *89*, 765–788.
- (3) Fricke, J.; Tillotson, T. Aerogels: Production, Characterization, and Applications. *Thin Solid Films* **1997**, *297*, 212–223.
- (4) Malek, K.; Coppens, M. O. Knudsen Self- and Fickian Diffusion in Rough Nanoporous Media. *J. Chem. Phys.* **2003**, *119*, 2801–2811.
- (5) Ganguly, S.; Brock, S. L. Toward Nanostructured Thermoelectrics: Synthesis and Characterization of Lead Telluride Gels and Aerogels. *J. Mater. Chem.* **2011**, *21*, 8800–8806.
- (6) Lan, T. W.; Su, K. H.; Chang, C. C.; Chen, C. L.; Ou, M. N.; Wu, D. Z.; Wu, P. M.; Su, C. Y.; Wu, M. K.; Chen, Y. Y. Enhancing the Figure of Merit in Thermoelectric Materials by Adding Silicate Aerogel. *Mater. Today Phys.* **2020**, *13*, 100215.
- (7) Gordon, M. P.; Zaia, E. W.; Zhou, P.; Russ, B.; Coates, N. E.; Sahu, A.; Urban, J. J. Soft PEDOT:PSS Aerogel Architectures for Thermoelectric Applications. *J. Appl. Polym. Sci.* **2017**, *134*, 44070.
- (8) Yanagishima, N.; Kanehashi, S.; Saito, H.; Ogino, K.; Shimomura, T. Thermoelectric Properties of PEDOT:PSS Aerogel Secondary-Doped in Supercritical CO₂ Atmosphere with Low Thermal Conductivity. *Polymer* **2020**, *206*, 122912.
- (9) Li, T. R.; Huang, Y. H.; Chen, J. X. M.; Sun, Y. C.; Aghababaei, O.; Saadatnia, Z.; Naguib, H. E. 3D Printing of Conductive Polymer Aerogel Thermoelectric Generator with Tertiary Doping. *Nano Energy* **2023**, *117*, 108909.
- (10) Khan, Z. U.; Edberg, J.; Hamed, M. M.; Gabrielson, R.; Granberg, H.; Wagberg, L.; Engquist, I.; Berggren, M.; Crispin, X. Thermoelectric Polymers and Their Elastic Aerogels. *Adv. Mater.* **2016**, *28*, 4556–4562.
- (11) Han, S. B.; Jiao, F.; Khan, Z. U.; Edberg, J.; Fabiano, S.; Crispin, X. Thermoelectric Polymer Aerogels for Pressure-Temperature Sensing Applications. *Adv. Funct. Mater.* **2017**, *27*, 1703549.
- (12) Sun, X.; Wei, Y.; Li, J.; Zhao, J.; Zhao, L.; Li, Q. Ultralight Conducting PEDOT:PSS/Carbon Nanotube Aerogels Doped with Silver for Thermoelectric Materials. *Sci. China. Mater.* **2017**, *60*, 159–166.
- (13) Wang, X. D.; Liu, P. P.; Jiang, Q. L.; Zhou, W. Q.; Xu, J. K.; Liu, J.; Jia, Y. H.; Duan, X. M.; Liu, Y. F.; Du, Y. K.; Jiang, F. X. Efficient DMSO-Vapor Annealing for Enhancing Thermoelectric Performance of PEDOT:PSS-Based Aerogel. *ACS Appl. Mater. Interface* **2019**, *11*, 2408–2417.
- (14) Kroon, R.; Ryan, J. D.; Kiefer, D.; Yu, L. Y.; Hynynen, J.; Olsson, E.; Muller, C. Bulk Doping of Millimeter-Thick Conjugated Polymer Foams for Plastic Thermoelectrics. *Adv. Funct. Mater.* **2017**, *27*, 1704183.
- (15) Okada, N.; Sato, K.; Yokoo, M.; Kodama, E.; Kanehashi, S.; Shimomura, T. Thermoelectric Properties of Poly(3-hexylthiophene) Nanofiber Aerogels with a Giant Seebeck Coefficient. *ACS Appl. Polym. Mater.* **2021**, *3*, 455–463.
- (16) Mano, G.; Murasawa, Y.; Shimamura, K.; Iso, A.; Kanehashi, S.; Shimomura, T. Fabrication, Characterization, and Thermoelectric Properties of Soft Polyurethane Foam Loaded with Semiconducting Poly(3-hexylthiophene) Nanofibers. *J. Appl. Polym. Sci.* **2022**, *139*, No. e52354.
- (17) Weinbach, Q.; Thakkar, S. V.; Carvalho, A.; Chaplais, G.; Combet, J.; Constantin, D.; Stein, N.; Collin, D.; Biniek, L. Efficient Control of a Mesoporous Fibrillar PEDOT:PSS Aerogel Structure for Promising Thermoelectric Applications. *Frontiers Electro. Mater.* **2022**, *2*, 875856.
- (18) Menon, R.; Yoon, C. O.; Moses, D.; Heeger, A. J. *Handbook of Conducting Polymer*, 2nd ed.; Skotheim, T. A., Elsenbaumer, R. L., Reynolds, J. R., Eds.; Marcel Dekker: New York, NY, 1998; Chapter 2, pp 27–84.
- (19) Kim, N.; Lee, B. H.; Choi, D.; Kim, G.; Kim, H.; Kim, J. R.; Lee, J.; Kahng, Y. H.; Lee, K. Role of Interchain Coupling in the Metallic State of Conducting Polymers. *Phys. Rev. Lett.* **2012**, *109*, 106405.
- (20) Kim, J. Y.; Jung, J. H.; Lee, D. E.; Joo, J. Enhancement of Electrical Conductivity of Poly(3,4-ethylenedioxythiophene)/Poly(4-styrenesulfonate) by a Change of Solvents. *Synth. Met.* **2002**, *126*, 311–316.
- (21) Bubnova, O.; Khan, Z. U.; Wang, H.; Braun, S.; Evans, D. R.; Fabretto, M.; Hojati-Talemi, P.; Dagnelund, D.; Arlin, J. B.; Geerts, Y. H.; Desbief, S.; Breiby, D. W.; Andreasen, J. W.; Lazzaroni, R.; Chen, W. M.; Zozoulenko, L.; Fahlman, M.; Murphy, P. J.; Berggren, M.; Crispin, X. Semi-Metallic Polymers. *Nat. Mater.* **2014**, *13*, 190–194.
- (22) Gueye, M. N.; Carella, A.; Massonnet, N.; Yvenou, E.; Brenet, S.; Faure-Vincent, J.; Pouget, S.; Rieutord, F.; Okuno, H.; Benayad, A.; Demadrille, R.; Simonato, J.-P. Structure and Dopant Engineering in PEDOT Thin Films: Practical Tools for a Dramatic Conductivity Enhancement. *Chem. Mater.* **2016**, *28*, 3462–3468.
- (23) Mantione, D.; del Agua, I.; Schaafsma, W.; ElMahmoudy, M.; Uguz, I.; Sanchez-Sanchez, A.; Sardon, H.; Castro, B.; Malliaras, G. G.; Mecerreyes, D. Low-Temperature Cross-Linking of PEDOT:PSS Films Using Divinylsulfone. *ACS Appl. Mater. Inter.* **2017**, *9*, 18254–18262.
- (24) Guex, A. G.; Puetzer, J. L.; Armgarth, A.; Littmann, E.; Stavrindou, E.; Giannelis, E. P.; Malliaras, G. G.; Stevens, M. M. Highly

Porous Scaffolds of PEDOT:PSS for Bone Tissue Engineering. *Acta Biomater.* **2017**, *62*, 91–101.

(25) Zhang, X. T.; Li, C. Y.; Luo, Y. J. Aligned/Unaligned Conducting Polymer Cryogels with Three-Dimensional Macroporous Architectures from Ice-Segregation-Induced Self-Assembly of PEDOT-PSS. *Langmuir* **2011**, *27*, 1915–1923.

(26) Friedel, B.; Keivanidis, P. E.; Brenner, T. J. K.; Abrusci, A.; McNeill, C. R.; Friend, R. H.; Greenham, N. C. Effects of Layer Thickness and Annealing of PEDOT:PSS Layers in Organic Photo-detectors. *Macromolecules* **2009**, *42*, 6741–6747.

(27) Ashby, M. F.; Evans, T.; Fleck, N. A.; Hutchinson, J. W.; Wadley, H. N. G.; Gibson, L. J. *Metal Foams: A Design Guide*, 1st ed.; Butterworth-Heinemann, 2000.

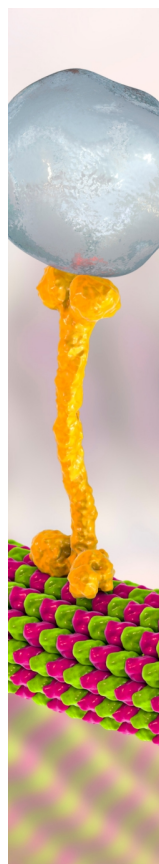
(28) Menon, R.; Yoon, C. O.; Moses, D.; Heeger, A. J. Metal-Insulator Transition in Doped Conducting Polymers. In *Handbook of Conducting Polymers*, 2nd ed.; Marcel Dekker, 1998; pp 27–84.

(29) Mott, N. F.; Jones, H. *The Theory of the Properties of Metals and Alloys*; Dover Publications, 1958.

(30) Seeger, K. *Semiconductor Physics: An Introduction*, 9th ed.; Springer, 2004.

(31) Fan, Z.; Ouyang, J. Y. Thermoelectric Properties of PEDOT:PSS. *Adv. Electron. Mater.* **2019**, *5*, 1800769.

(32) Kim, G. H.; Shao, L.; Zhang, K.; Pipe, K. P. Engineered Doping of Organic Semiconductors for Enhanced Thermoelectric Efficiency. *Nat. Mater.* **2013**, *12*, 719–723.



CAS BIOFINDER DISCOVERY PLATFORM™

BRIDGE BIOLOGY AND CHEMISTRY FOR FASTER ANSWERS

Analyze target relationships,
compound effects, and disease
pathways

Explore the platform

CAS
A Division of the
American Chemical Society